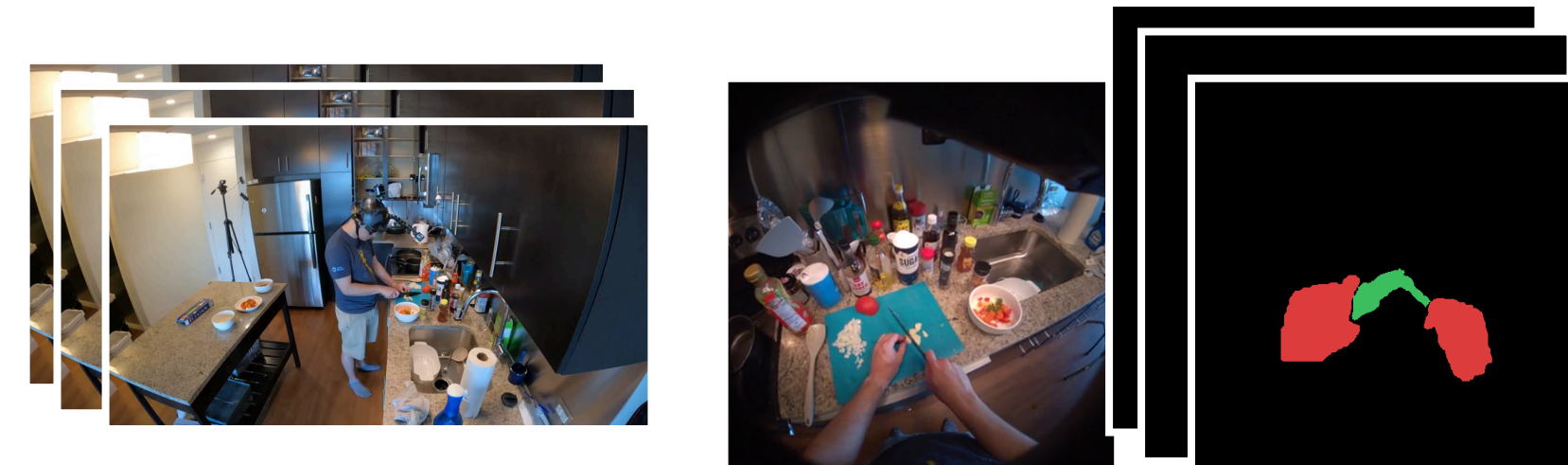


Training inputs

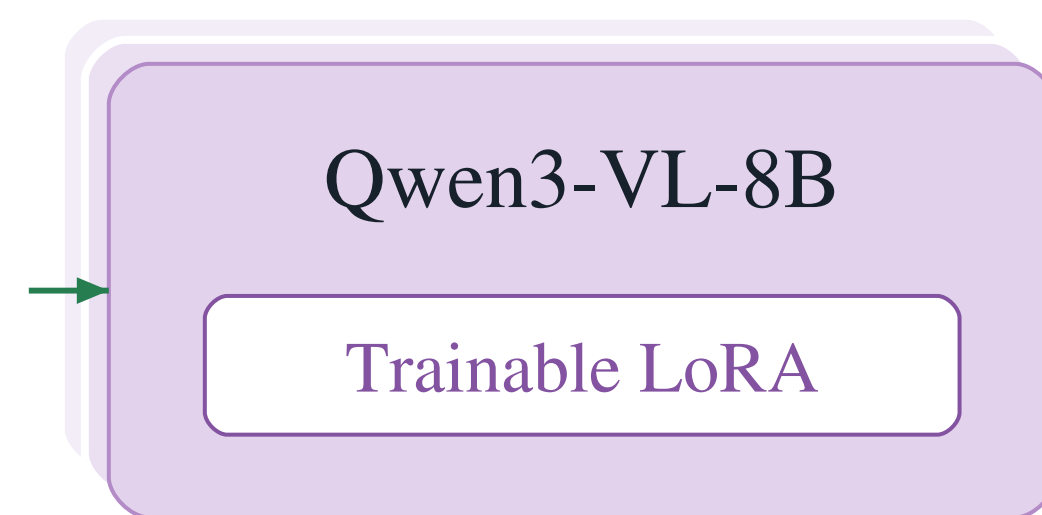


Exo video
Narration + viewpoint instruction

Ego first frame

Ego HOI M

Stage 1 · Difference-weighted SFT

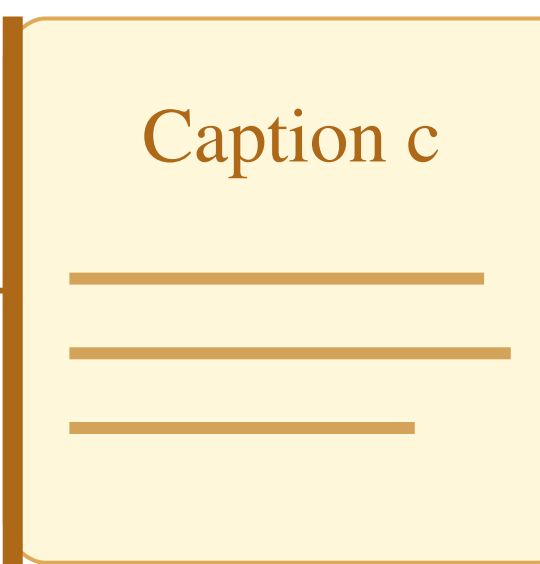
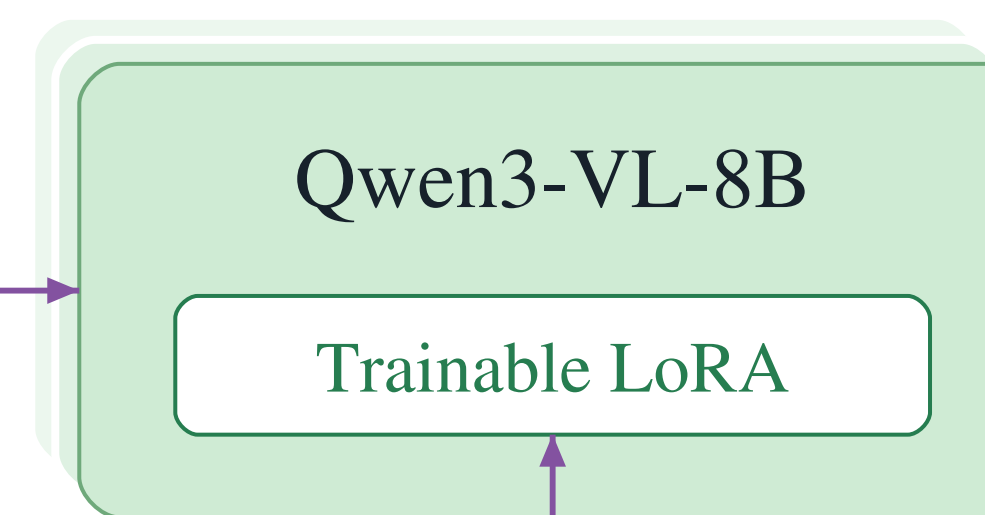


Teacher target T · blind baseline B

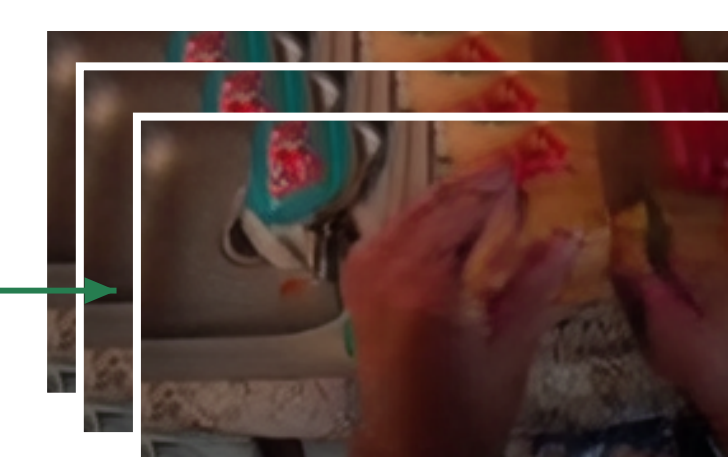
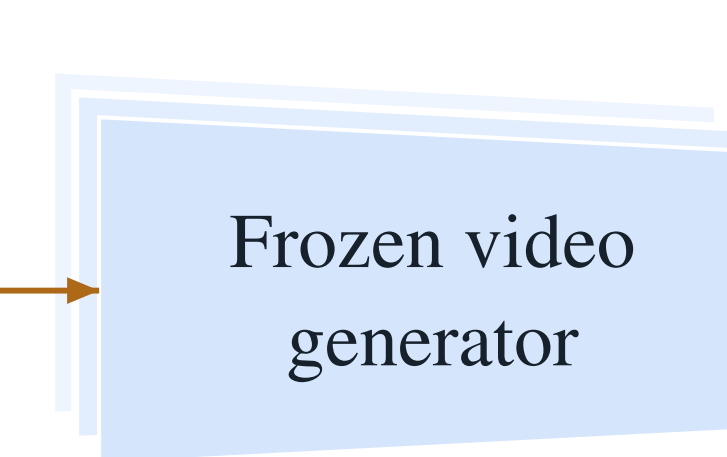
$$\mathcal{L}_{\text{SFT}} \propto - \sum_t w_t \log \pi_{\theta}(T_t | u, T_{<t})$$

w_t : 2.5 net-new / 1.0 partial / 0.3 redundant

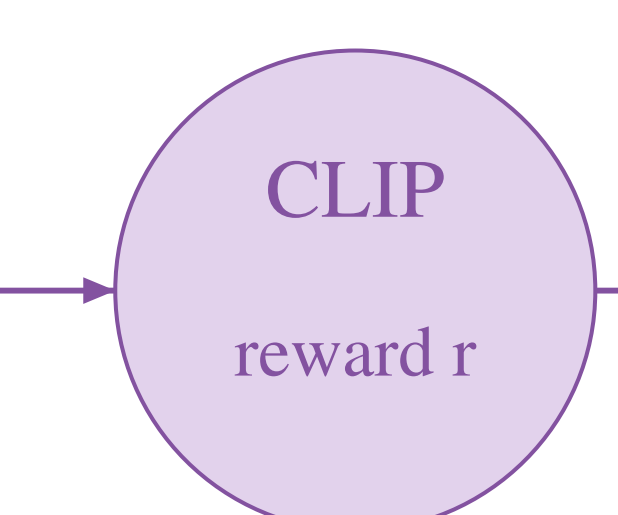
Same inputs u



Candidate caption



HOI crop



Crop × net-new text

$$r = \langle \bar{e}_{\text{img}}(\text{Crop}_M(G(c))), \bar{e}_{\text{txt}}(\Delta_{BC}) \rangle$$

$$\mathcal{L}_{\text{GRPO}} = - \langle \min(\rho A, \text{clip}(\rho, 1 - \delta, 1 + \delta)A) \rangle + \beta D_{\text{KL}}$$

GRPO update · LoRA only

Stage 2 · Generation-grounded GRPO

Reward signal